

From Probabilistic Population Protocols to Non-Autocatalytic Reactions: An Informal Walkthrough

2026-04-24

Abstract

This note walks through the proof of BD Theorem 11.2 (PP $\rightarrow$ NAP) in ordinary mathematical prose. No formal-verification language appears. The goal is for the reader to be able to evaluate the mathematical content on its own terms, step by step, before consulting any machine artifact.

1 What we want to prove

A *probabilistic population protocol* (PP) on n species is specified by a tensor of non-negative rationals

$$c_{r,i,j} \in \mathbb{Q}_{\geq 0}, \quad r, i, j \in \{1, \dots, n\},$$

symmetric in (i, j) and subject to the mass-conservation constraint

$$\sum_{r=1}^n c_{r,i,j} = 2 \quad \text{for every pair } (i, j). \quad (\text{P1})$$

The reading is: when one copy of species i collides with one copy of species j , they react into a pair of output copies, with $c_{r,i,j}/2$ being the expected number of r -copies produced. Condition (P1) records that two input copies give exactly two output copies (mass is conserved).

On the probability simplex $\Delta^{n-1} = \{x \in \mathbb{R}_{\geq 0}^n : \sum_k x_k = 1\}$, the associated mean-field ODE is

$$\dot{x}_r = \sum_{i,j} c_{r,i,j} x_i x_j - 2x_r. \quad (\text{P2})$$

The quadratic term is the *production* rate of r ; the $-2x_r$ is the *degradation* rate, which integrates the fact that whenever any reaction happens r loses a copy proportional to its concentration.

A *non-autocatalytic reaction* (NAP) in the chemical sense is an elementary reaction $I \rightarrow O$ where neither I nor O dominates the other by pure replication of the same species. Formally, we will work degree-3: encode states as cubic multinomials $v_\alpha = x^\alpha$ with $|\alpha| = 3$, and ask that the chain-rule rate of each $\dot{v}_\alpha$ factor into a sum of transitions $\beta \rightarrow \gamma$ of cubed exponents where $\beta \neq \alpha$ and $\gamma \neq \alpha$ (“no species just catalyzes itself”). The precise condition is the *balanced non-autocatalytic split* defined in §4.

The claim. Every PP can be implemented by a collection of NAPs in cubed coordinates, after adding one slack species.

2 Step 1: folding degradation into a signed quadratic field

Define the signed coefficients

$$q_{r,i,j} := c_{r,i,j} - \llbracket i = r \rrbracket - \llbracket j = r \rrbracket, \quad (1)$$

where $\llbracket \cdot \rrbracket$ denotes the Iverson bracket. Using $\sum_k x_k = 1$ twice to rewrite $2x_r = 2x_r \sum_k x_k = \sum_{i,j} (\llbracket i = r \rrbracket + \llbracket j = r \rrbracket) x_i x_j$, we get on the simplex

$$\dot{x}_r = \sum_{i,j} q_{r,i,j} x_i x_j. \quad (\text{P3})$$

The signed tensor q inherits the conservation law

$$\sum_r q_{r,i,j} = \left(\sum_r c_{r,i,j} \right) - 2 = 0. \quad (\text{P4})$$

So far (P3)–(P4) is a lossless rewriting of the PP. Call any tensor $q_{r,i,j}$ satisfying (P4) a *quadratic conservative field* (QCF). Every PP is a QCF by folding; not every QCF comes from a PP. The passage from PP to QCF is a purely algebraic book-keeping step.

Reading of the signs. A positive entry $q_{r,i,j} > 0$ means that monomial $x_i x_j$ contributes a net *production* of r . A negative entry means the corresponding monomial is a net *consumer* of r — but notice that by construction these negative entries come only from the folding terms $-\llbracket i = r \rrbracket - \llbracket j = r \rrbracket$, which are not free parameters: they encode “whenever i reacts with j , the species i and the species j are each spent once”. The magnitude of a consumer entry is tied by (P4) to the production entries in its own column.

Lemma 1 (no positive self-self on the diagonal). *For any PP, the folded tensor satisfies*

$$q_{r,r,r} \leq 0 \quad \text{for every } r.$$

Proof. From $c_{s,r,r} \geq 0$ and $\sum_s c_{s,r,r} = 2$ one has $c_{r,r,r} \leq \sum_s c_{s,r,r} = 2$. Therefore $q_{r,r,r} = c_{r,r,r} - 2 \leq 0$. $\square$

We will denote this property by (NSS). It is automatic for any PP.

3 Step 2: the λ -lift — carving out a slack species

Pick a species j_0 and a parameter $\lambda \in (0, 1)$. Introduce one new species, which we call 0 (the *slack*), and rename

$$u := \lambda x_{j_0}, \quad r_0 := (1 - \lambda) x_{j_0}, \quad u + r_0 = x_{j_0}.$$

For every other species $i \neq j_0$, keep $y_i := x_i$. The enlarged state is $(r_0, y_1, \dots, y_{j_0-1}, u, y_{j_0+1}, \dots, y_n)$. On the simplex, $\sum_i y_i + u + r_0 = 1$. The scalar $\alpha_i := \lambda$ if $i = j_0$ else 1 records the local rescaling.

Goal of the lift. We want the enlarged field to satisfy two regularity properties:

(NSM) *No Self-Mix on slack:* $\tilde{q}_{r,0,k} = 0$ and $\tilde{q}_{r,k,0} = 0$ for every r and k . In words: the slack species 0 never appears as a reactant factor.

(NSS) $\tilde{q}_{r,r,r} \leq 0$ for every r in the enlarged index set.

Their conjunction we call *slack-structured*. The slack species' role is purely to absorb stoichiometric excess: it will be consumed or produced but never participate *quadratically* as a reactant with itself or with anyone else.

Lift formula. Substitute $x_{j_0} = (u + r_0)/\lambda \cdot \lambda = u + r_0$ and enforce the slack dynamics $\dot{r}_0 = \dot{x}_{j_0} - \dot{u} = (1 - \lambda)\dot{x}_{j_0}$. The signed coefficient in the enlarged tensor is

$$\tilde{q}_{r', i', k'} = \begin{cases} 0 & \text{if } i' = 0 \text{ or } k' = 0, \\ \frac{(1 - \lambda) q_{j_0, I, K}}{\alpha_I \alpha_K} & \text{if } r' = 0, i' = I^+, k' = K^+, \\ \frac{\alpha_R q_{R, I, K}}{\alpha_I \alpha_K} & \text{if } r' = R^+, i' = I^+, k' = K^+, \end{cases} \quad (\Lambda)$$

where R^+, I^+, K^+ indicate the image of the old species indices in the enlarged set (slot 0 is slack; old index j goes to $j^+ = j + 1$ abstractly). The rescaling $1/(\alpha_I \alpha_K)$ compensates for the fact that an old monomial $x_I x_K$ becomes, after substitution, $u \cdot x_K / \lambda = \alpha_I \alpha_K \cdot \dots$; the upshot is that the vector field $(\dot{\cdot})$ is preserved.

Lemma 2 ($\tilde{q}$ is still conservative). *For every (i', k') , $\sum_{r'} \tilde{q}_{r', i', k'} = 0$.*

Proof. For $i' = 0$ or $k' = 0$ every summand is zero. For $i' = I^+, k' = K^+$, summing the $r' = 0$ row and the $r' = R^+$ rows:

$$\alpha_I \alpha_K \sum_{r'} \tilde{q}_{r', I^+, K^+} = (1 - \lambda) q_{j_0, I, K} + \sum_R \alpha_R q_{R, I, K}.$$

Use $\alpha_R = 1$ for $R \neq j_0$ and $\alpha_{j_0} = \lambda$ to break off the $R = j_0$ term:

$$= (1 - \lambda) q_{j_0, I, K} + \sum_R q_{R, I, K} + (\lambda - 1) q_{j_0, I, K} = \sum_R q_{R, I, K} = 0,$$

using (P4) for the original q . □

Lemma 3 ($\tilde{q}$ is slack-structured). *If the original q satisfies (NSS), and $\lambda > 0$, then $\tilde{q}$ satisfies both (NSM) and (NSS).*

Proof. (NSM) is immediate from the first line of (Λ) . For (NSS) on the slack row, $\tilde{q}_{0,0,0} = 0 \leq 0$. For a non-slack row $r' = R^+$:

$$\tilde{q}_{R^+, R^+, R^+} = \frac{\alpha_R q_{R, R, R}}{\alpha_R^2} = \frac{q_{R, R, R}}{\alpha_R} \leq 0,$$

since $q_{R, R, R} \leq 0$ by hypothesis and $\alpha_R > 0$. □

Remark 4 (why we also need $\lambda < 1$). *The algebraic lemma uses only $\lambda > 0$. The hypothesis $\lambda < 1$ is used downstream for the simplex interpretation: the slack coordinate $r_0 = (1 - \lambda)x_{j_0}$ must be non-negative, so $1 - \lambda \geq 0$, and strict positivity $1 - \lambda > 0$ keeps the slack non-trivial.*

At the end of this step we have, from any PP, a QCF on $n + 1$ species (slot 0 is slack) that satisfies slack-structured = (NSM) + (NSS). This is the field we actually split.

4 Step 3: the r^2 -trick and cubic coordinates

3a. r^2 -trick: adding a slack factor to every production

Multiply every production monomial by x_0^2 . Concretely, replace

$$\tilde{q}_{r,a,b} x_a x_b \longrightarrow \tilde{q}_{r,a,b} x_0^2 x_a x_b.$$

On the simplex, multiplying by $x_0^2 \leq 1$ is a pure time rescaling; the qualitative dynamics are unaffected. The payoff is that every production monomial now carries *two* guaranteed slack factors. Writing $\mu_{0,a,b}(k) := 2\llbracket k = 0 \rrbracket + \llbracket k = a \rrbracket + \llbracket k = b \rrbracket$ for the exponent vector of $x_0^2 x_a x_b$, this is the monomial that will be split.

3b. Cubing: moving to degree-3 coordinates

For an exponent vector $\alpha : \{0, 1, \dots, n\} \rightarrow \mathbb{N}$ with $|\alpha| = 3$, let $v_\alpha := c_\alpha x^\alpha$ (multinomial coefficient). The chain rule gives

$$\dot{v}_\alpha = \sum_s \alpha(s) c_\alpha \dot{x}_s x^{\alpha - e_s},$$

where e_s is the s -th standard basis vector. Substituting $\dot{x}_s = \sum_{a,b} \tilde{q}_{s,a,b} x_0^2 x_a x_b$ (post r^2 -trick), each positive entry $\tilde{q}_{s,a,b} > 0$ contributes a degree-6 positive monomial to $\dot{v}_\alpha$, with exponent

$$\delta(k) := (\alpha(k) - \llbracket k = s \rrbracket) + \mu_{0,a,b}(k). \quad (\delta)$$

Provided $\alpha(s) > 0$, we have $\alpha(k) - \llbracket k = s \rrbracket \geq 0$ and the expression is well-defined.

3c. The key combinatorial notion — the split

Given δ as above, a *balanced split* is a pair of exponent vectors (β, γ) with

$$\beta + \gamma = \delta, \quad |\beta| = |\gamma| = 3.$$

A balanced split is *non-autocatalytic* for α if

$$\beta \neq \alpha \quad \text{and} \quad \gamma \neq \alpha.$$

The interpretation is a cubic reaction $\beta \rightarrow \gamma$ whose reactant and product triples are both different from α — so α genuinely transitions and is not merely being catalyzed.

Why this definition? We want a splitting that corresponds to a reaction in which no α -monomer is present on *both* sides. If $\beta = \alpha$, the reaction is $\alpha \rightarrow \gamma$ with α present on the reactant side in full; but $\delta - \alpha = \gamma$, meaning one copy of α just sits around unchanged during the reaction. That is the classic self-catalysis pattern we are excluding.

5 Step 4: existence of a non-autocatalytic split, per pair

This is where the work is. The theorem we prove says: whenever $\tilde{q}_{s,a,b} > 0$ with the slack structure in place, a balanced non-autocatalytic split of δ exists.

Theorem 5 (per-pair splitter). *Let $\tilde{q}$ be slack-structured on species $\{0, 1, \dots, n\}$, and let α be a cubic exponent vector. Let s, a, b be species with $\alpha(s) > 0$, $a \neq 0$, $b \neq 0$ (so that $\tilde{q}_{s,a,b}$ is a positive non-slack-reactant entry by (NSM)). Suppose*

$$s \neq 0 \implies a \neq s \text{ or } b \neq s. \quad (\star)$$

Then δ (defined by (δ)) admits a balanced non-autocatalytic split.

We split the argument into three cases based on the identity of s and the location of α 's mass.

Case A: $s = 0$ (slack is the source). Then the $-[k = s]$ term in (δ) subtracts a slack unit, and $\mu_{0,a,b}$ adds two. Net effect: δ has at least one slack unit and at least one unit each on a and b (which are non-slack by hypothesis). So the support of δ contains $\{0, a, b\}$ with multiplicities ≥ 1 everywhere and the total is $|\delta| = 6$.

We split by:

$$\beta := e_0 + e_a + e_t, \quad \gamma := e_0 + e_b + e_{t'},$$

where $e_t, e_{t'}$ are chosen to make $\beta + \gamma = \delta$ and to avoid equality with α . Whenever $\delta - e_0 - e_a - e_b$ has support $\{0\}$ (i.e. δ concentrates on slack and $\{a, b\}$), β, γ trivially differ from α because $\alpha(0)$ would have to absorb all three mass units of α , but α also has $\alpha(s) = \alpha(0) > 0$ and two more units distributed somewhere. A case distinction on whether those remaining units fall on a, b , or a third species completes the construction; two symmetric branches (“distinct” and “doubled”) handle the sub-cases.

In all sub-cases $\beta \neq \alpha$ and $\gamma \neq \alpha$ because each contains one slack unit, whereas α has at most 3 slack units, and the remaining structure differs.

Case B: $s \neq 0$ and $a \neq s$. This is the central case. The slack 0 in δ enters with multiplicity 2 (from $\mu_{0,a,b}$). The source index s enters with multiplicity $\alpha(s) - 1 \geq 0$. The species a enters with multiplicity $\alpha(a) + 1 \geq 1$.

The *foreign pair* is $(0, a)$: neither coordinate equals s . We set

$$\beta := e_0 + e_a + \eta_\beta, \quad \gamma := e_0 + e_\gamma + \eta_\gamma,$$

where the concrete shapes of η_β, η_γ are adjusted so that $\beta + \gamma = \delta$ and the total on each side is 3.

Why is $\beta \neq \alpha$? Because $\alpha(s) > 0$, if we had $\beta = \alpha$, then the coordinate s in β would need to equal $\alpha(s) \geq 1$. But β only puts mass on $\{0, a, \text{one more species}\}$, none of which is s (since $a \neq s$, and any third species we can always pick to differ from s). Contradiction. Symmetrically for γ .

The construction of the third species filling the remaining slot is a combinatorial flow-matching step. The key observation is that we have six units of mass in δ (two on slack, then the remainder), and we need to split them into two halves of three, each avoiding α . This is a degree-3 bin-packing problem with finite feasibility; a direct case analysis on the support sizes $|\text{supp}(\alpha)| \in \{1, 2, 3\}$ shows a solution always exists. (This is the “flow-matching lemma”, which we sketch in §9 below.)

Case C: $s \neq 0$ and $b \neq s$. Symmetric to Case B, with foreign pair $(0, b)$.

The disjunction $(\star)$ ensures that in the $s \neq 0$ branch at least one of $a \neq s$, $b \neq s$ holds, so either Case B or Case C applies.

Lemma 6 $(\star)$ is automatic under slack structure). *If $\tilde{q}$ satisfies (NSS), $\tilde{q}_{s,a,b} > 0$, and $s \neq 0$, then $a \neq s$ or $b \neq s$.*

Proof. Suppose both $a = s$ and $b = s$. Then $\tilde{q}_{s,a,b} = \tilde{q}_{s,s,s} \leq 0$ by (NSS), contradicting positivity. $\square$

Combining Theorem 5 with Lemma 6:

Corollary 7 (main technical result). *If $\tilde{q}$ is a slack-structured QCF, then for every cubed α and every positive entry $\tilde{q}_{s,a,b} > 0$ with $a, b \neq 0$ and $\alpha(s) > 0$, the monomial δ admits a balanced non-autocatalytic split.*

6 Step 5: putting it together — PP to NAP

Theorem 8 (PP $\rightarrow$ NAP). *Let $c_{r,i,j}$ be a PP on n species. Fix $j_0 \in \{1, \dots, n\}$ and $\lambda \in (0, 1)$, and form the λ -lift $\tilde{q}$ of the folded tensor $q_{r,i,j} = c_{r,i,j} - \llbracket i = r \rrbracket - \llbracket j = r \rrbracket$ according to (Λ) . Then for every cubed exponent vector α on $n + 1$ species, every source s with $\alpha(s) > 0$, and every pair (a, b) with $a, b \neq 0$ and $\tilde{q}_{s,a,b} > 0$: the associated degree-6 monomial δ (after the r^2 -trick) admits a balanced non-autocatalytic split of α .*

Proof. Compose the three steps:

1. $c \geq 0$, $\sum_r c_{r,i,j} = 2 \Rightarrow q$ satisfies (NSS) (Lemma 1).
2. q is (NSS) and $\lambda \in (0, 1) \Rightarrow \tilde{q}$ is slack-structured on $n + 1$ species (Lemmas 2 and 3).
3. $\tilde{q}$ is slack-structured $\Rightarrow$ for every positive entry $\tilde{q}_{s,a,b} > 0$ with $a, b \neq 0$, δ admits a balanced non-autocatalytic split for every α with $\alpha(s) > 0$ (Corollary 7).

$\square$

7 Aggregation and the reading as a CRN

Theorem 8 produces a split for each positive (s, a, b) entry of $\tilde{q}$ and each cubed α with $\alpha(s) > 0$. The protocol-level picture follows by summing over these:

For every cubed α and every source s with $\alpha(s) > 0$, the r^2 -lifted chain-rule contribution to $\dot{v}_\alpha$,

$$\alpha(s) \sum_{a,b} \tilde{q}_{s,a,b} x^\delta,$$

routes through a weighted sum of cubic reactions $\beta \rightarrow \gamma$, none of which has α on either side.

The routing uses a simple identity: for each positive entry, the rate $\alpha(s) \tilde{q}_{s,a,b}$ is assigned to the split reaction $\beta \rightarrow \gamma$ found by Theorem 5. The sum over (a, b) with $\tilde{q}_{s,a,b} > 0$ is the total positive contribution.

What about negative entries? A negative entry $\tilde{q}_{s,a,b} < 0$ corresponds, in the PP picture, to the folding terms $-\llbracket i = r \rrbracket - \llbracket j = r \rrbracket$: these are *degradation* edges attached to the reaction “ $(i, j) \rightarrow$ (something)”. They are not independent free variables but consumption terms coupled to the same reaction via (P4). In the CRN translation, they are realized as the loss of one copy of i (or j) upon firing, which is structurally non-autocatalytic: destroying one copy of a species is obviously not self-replication.

8 Why the forbidden-edge concern does not bite

A natural worry is the following. Suppose a monomial $-kuv$ shows up in $\dot{v}$ (negative coefficient, large k). The same monomial uv can also appear with a positive coefficient K in some $\dot{w}$. If k is large enough, wouldn't removing the forbidden edge $(-k)$ destroy the feasibility of a global flow condition and invalidate the split?

The answer is that no such global flow condition appears in our argument. Three layers of structural isolation rule it out:

First layer: folding-derived negatives are reaction bookkeeping, not free edges. The only negatives appearing in our QCF are of the form $-\llbracket i = r \rrbracket - \llbracket j = r \rrbracket$, which record “when i and j react, one copy of i is consumed, and so is one copy of j ”. This is the same reaction already accounted for on the positive side, not a free edge to match independently.

Second layer: the coefficients of a single monomial across different $\dot{x}_r$ are tied by (P1). In the original PP, every monomial $x_i x_j$ corresponds to one reaction $i + j \rightarrow$ products, with the distribution over product species governed by $c_{r,i,j}/2$ and $\sum_r c_{r,i,j} = 2$. The “ $-k$ ” and “ $+K$ ” the worry imagines are not independent parameters: they are both extracted from the *same* probability vector $c_{\cdot,u,v}$. You cannot have “large k ” without forcing the corresponding c 's to be small; the coupling is through (P1). Thus a large forbidden edge *cannot* be created in isolation of its positive counterparts.

Third layer: the splitting is per pair, not global. The per-pair theorem (Theorem 5) takes one (s, a, b) at a time and produces a split using only *that entry's* positivity and the slack structure. There is no cross- (s, a, b) flow feasibility that has to be solved. The r^2 -trick contributes one slack factor of multiplicity 2 per entry; the original PP contributes at least one non-source reactant (from (NSS) via Lemma 6); these are enough to guarantee the foreign-pair condition $(\star)$ independently for each entry. Aggregation over all entries is then a linear sum — the rates for each positive entry add up — with no constraint from the negatives.

In short: the proof never asks “is there a feasible flow routing all positive and negative edges?” It asks “given each positive edge, can I reshape its cubic footprint into a non-autocatalytic transition?” The answer is yes, one edge at a time; the negatives are already bookkeeping on the same reaction.

9 The flow-matching lemma

For completeness we sketch the degree-3 existence claim used inside Case B and Case C of Theorem 5.

Lemma 9 (split existence given foreign pair). *Let $\delta : \{0, 1, \dots, n\} \rightarrow \mathbb{N}$ with $|\delta| = 6$. Let α be a cubic exponent vector. Suppose there exist two distinct indices $i_1, i_2 \in \text{supp}(\delta)$ with*

$$i_1 \neq s, \quad i_2 \neq s, \quad \delta(i_1) \geq 1, \quad \delta(i_2) \geq 1,$$

where s is a fixed “source” species with $\alpha(s) > 0$. Then there is a balanced split (β, γ) of δ with $\beta \neq \alpha$ and $\gamma \neq \alpha$.

Proof sketch. Start by placing $e_{i_1} + e_{i_2}$ into β ; this already ensures $\beta \neq \alpha$ because β now contains at least one of $i_1, i_2 \neq s$, whereas the triple α (with $\alpha(s) \geq 1$) has at least one unit at s that β lacks.

The remaining mass is $\delta - e_{i_1} - e_{i_2}$, of total 4. We need to add one more unit to β (to reach size 3) and the rest to γ (of size 3). Choose the single unit for β so that $\gamma = \delta - \beta$ also differs from α . Two sub-cases:

- If $\delta - e_{i_1} - e_{i_2}$ places mass on a species $t \neq s$, take that t to complete β ; then γ 's remaining mass on s is $\delta(s) = \alpha(s)$ minus whatever adjustments — and the supports disagree with α because γ 's third unit is at some $t' \neq s$ (by residual mass count, γ always has at least one non- s unit).
- If the remaining mass lives entirely on s , then we use the fact that $|\text{supp}(\alpha)| \leq 3$ but γ 's support of size 3 then concentrates at s with multiplicity 3, which forces $\alpha = 3e_s$ for equality. In that case we re-route one unit from β to γ and rebalance (swapping one unit between β and γ) — again possible because at least two distinct off- s indices are available.

In each sub-case explicit formulas for (β, γ) can be written down; the verification that $\beta, \gamma \neq \alpha$ is a one-line count of multiplicities. $\square$

Remark 10. *The r^2 -trick guarantees $\delta(0) \geq 2$, so the slack 0 can always serve as one of the indices i_1, i_2 . The other index comes from the PP reactant distinction: if $s \neq 0$, the non-autocatalytic hypothesis $(\star)$ yields either $a \neq s$ or $b \neq s$, providing the second foreign index.*

10 Summary

The argument is modular. Three independent steps compose into the main theorem:

- Step 1.** Fold the PP into a signed conservative field q satisfying (NSS).
- Step 2.** λ -lift into $\tilde{q}$ on one extra slack species, preserving (NSS) and gaining (NSM).
- Step 3.** r^2 -trick and cubing: every positive entry of $\tilde{q}$ together with any cubic α with non-zero source mass produces a degree-6 footprint δ admitting a balanced non-autocatalytic split.

No step requires a global flow-feasibility check; each entry is handled in isolation. The negatives introduced by folding are reaction bookkeeping, not free edges, and they are tied by (P1) to the positive entries they accompany. This is why the “forbidden edge with large weight” scenario does not break the construction.